Geometric and Negative Binomial distributions

Geometric distribution $X \sim Geom(p)$

It is used to model the number of independent trials until the first "success" happens.

Definition

We say $X \sim Geom(p)$, then
X: = number of trials untill first success happens (we also count the trail in which success happens)
X can take values, $1, 2, 3, 4, \cdots$

$$P(X = k) = p(1-p)^{k-1};\ \text{where } k = 1, 2, 3, \cdots$$
$$E[X] = \frac{1}{p}$$
$$Var(X) = \frac{1-p}{p^2}$$

Eg 3. A slot machine at a casino randomly rewards 15% of the attempts. Assume that all attempts are independent.

(a) What is the probability that your first reward occurs on your fourth trial?

X = number of trials untill first reward

$p(\text{reward}) = 0.15 \implies X \sim \text{Geom}(0.15)$

$$p(X=4) = (0.15)(0.85)^{4-1}$$
$$= (0.15)(0.85)^{3}$$

Eg 3. A slot machine at a casino randomly rewards 15% of the attempts. Assume that all attempts are independent.

(b) What is the probability that your first reward occurs on your seventh trial?

X = number of trials untill first reward

$p(\text{reward}) = 0.15 \implies X \sim Geom(0.15)$

$$p(X = 7) = (0.15)(0.85)^{7-1}$$

$$= (0.15)(0.85)^{6}$$

Eg 3. A slot machine at a casino randomly rewards 15% of the attempts. Assume that all attempts are independent.

(c) What is the probability that your get three rewards in ten trials?

Y = number of rewards (success) in ten trials

$Y \sim Bin(n=10, p=0.15)$

$$P(Y=3) = \binom{10}{3} p^3 (1-p)^7$$
$$= \binom{10}{3} (0.15)^3 (0.85)^7$$

Eg 3. A slot machine at a casino randomly rewards 15% of the attempts. Assume that all attempts are independent.

(d) What is the probability that your third reward occurs on your tenth trial?

$p =$ probability of success in one trial $= 0.15$.

$r = 3$

$X :=$ number of independent trials untill 3 successes occur

$X \sim NBin(r = 3, p = 0.15)$

$$P(X = 10) = \binom{9}{2} p^3 (1-p)^7 = \binom{9}{2} (0.15)^3 (0.85)^7$$

Negative Binomial distribution $X \sim NBin(r, p)$

It is used to model the number of independent trials until r "success" happens.

Definition

We say $X \sim NBin(r, p)$, then
X: = number of trails until r success occurs
X can take values, $r, r+1, r+2, \cdots, \infty$

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}$$

$$E[X] = \frac{r}{p}$$

$$Var(X) = \frac{r(1-p)}{p^2}$$

Eg 3. A slot machine at a casino randomly rewards 15% of the attempts. Assume that all attempts are independent.

(e) What is the probability that the forth reward occurs on the fifteen trials?

$p =$ probability of success in one trial $= 0.15.$

$r = 4$

$X :=$ number of independent trial untill 4 success occurs

$X \sim NBin(r=4, p=0.15)$

$$p(X=15) = \binom{14}{3} p^4 (1-p)^{11} = \binom{14}{3} (0.15)^4 (0.85)^{11}$$